

ADVANCED LINEAR ALGEBRA

HOMEWORK 1

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P.1.2 Is $[1 \ 0 \ 1]^\top \in \mathbb{R}^3$ in the span of $[-1 \ 3 \ 2]^\top$ and $[1 \ 1 \ -2]^\top$? Why?

Solution.

□

P.1.3 Is $[7 \ 8 \ 9]^\top \in \mathbb{R}^3$ in the span of $[1 \ 2 \ 3]^\top$ and $[4 \ 5 \ 6]^\top$? Why? Is $[10 \ 11 \ 12]^\top$?

Solution.

□

P.1.8 Consider the following lists of vectors in $\mathbb{R}^3$:

$$\beta = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ s \\ 6 \end{bmatrix}, \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix},$$

and

$$\gamma = \begin{bmatrix} 4 \\ 2 \\ 6 \end{bmatrix}, \begin{bmatrix} 1 \\ t \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix},$$

For what values of s is β linearly dependent. For what values of t is γ linearly dependent.

Solution.

□

P.1.12 Show that the set of $n \times n$ Hermitian matrices is a real vector space.

Proof.

□

P.1.13 Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

(a) Choose two of the following and show that each is a factorization $A = XY$, in which the columns of X and rows of Y are each linearly independent:

$$\begin{bmatrix} 1 & 2 \\ 4 & 5 \\ 7 & 8 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}, \begin{bmatrix} 2 & 3 \\ 5 & 6 \\ 8 & 9 \end{bmatrix} \begin{bmatrix} 2 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 3 \\ 4 & 6 \\ 7 & 9 \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 1 \end{bmatrix}, \begin{bmatrix} -1 & 3 \\ -1 & 9 \\ -1 & 15 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & -\frac{3}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

(b) Which of the decompositions is the one described by Theorem 1.7.5? Why?

Solution.

□

P.1.27 (a) Show that $\text{col}(A) = \mathbb{R}^3$ for

$$A = [\mathbf{a}_1 \mathbf{a}_2 \mathbf{a}_3 \mathbf{a}_4] = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \in \mathbf{M}_n(\mathbb{R})$$

(b) Find $i, j, k \in 1, 2, 3, 4$ such that $\text{span}\{\mathbf{a}_i, \mathbf{a}_j, \mathbf{a}_k\} = \mathbb{R}^3$. And find distinct $i, j, k \in 1, 2, 3, 4$ such that $\text{span}\{\mathbf{a}_i, \mathbf{a}_j, \mathbf{a}_k\}$.

Solution.

□